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Set 18: Product Inventory Management

Data Visualization using R

Aim

To analyse product inventory using a bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the inventory data as a data frame. The data has no category column, so categories are assumed.
3. Question 1: draw the bar chart of quantity with `geom_col()`.
4. Question 2: draw the stacked bar chart of quantity by category with `geom_col()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset (Category is assumed, as the data has no category column)
inv <- data.frame(Product_ID = 1:3, Product = c("Product A", "Product B", "Product C"),
                  Quantity = c(250, 175, 300), Price = c(20, 15, 18),
                  Category = c("Category 1", "Category 2", "Category 1"))
print(inv)
print(aggregate(Quantity ~ Category, inv, sum))

# 1. Bar chart - quantity of each product
p1 <- ggplot(inv, aes(Product, Quantity)) + geom_col(fill = "steelblue") +
  geom_text(aes(label = Quantity), vjust = -0.5) +
  labs(title = "Quantity of Each Product", x = "Product", y = "Quantity Available")

# 2. Stacked bar chart - products within categories
p2 <- ggplot(inv, aes(Category, Quantity, fill = Product)) + geom_col() +
  labs(title = "Quantity by Category", x = "Category", y = "Quantity")

# 3. Table
tb <- tableGrob(inv[, 1:4], rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Inventory Dashboard")

ggsave("set18_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set18_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set18_q3.png", arrangeGrob(tb, top = "Product Inventory Table"), width = 5, height = 2, bg = "white")
ggsave("set18_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

Product_ID	Product	Quantity	Price	Category
1	1 Product A	250	20	Category 1
2	2 Product B	175	15	Category 2
3	3 Product C	300	18	Category 1
Category Quantity				
1	Category 1	550		
2	Category 2	175		

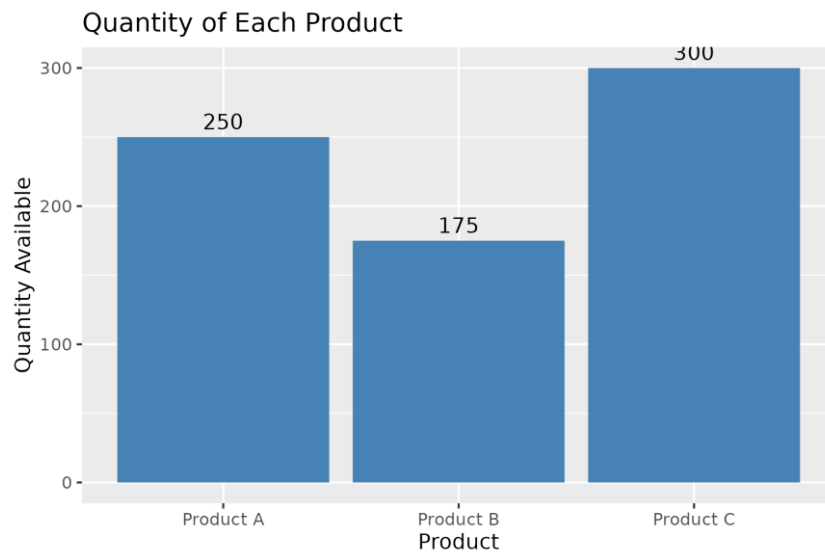


Figure 18.1: Q1 - Quantity of each product

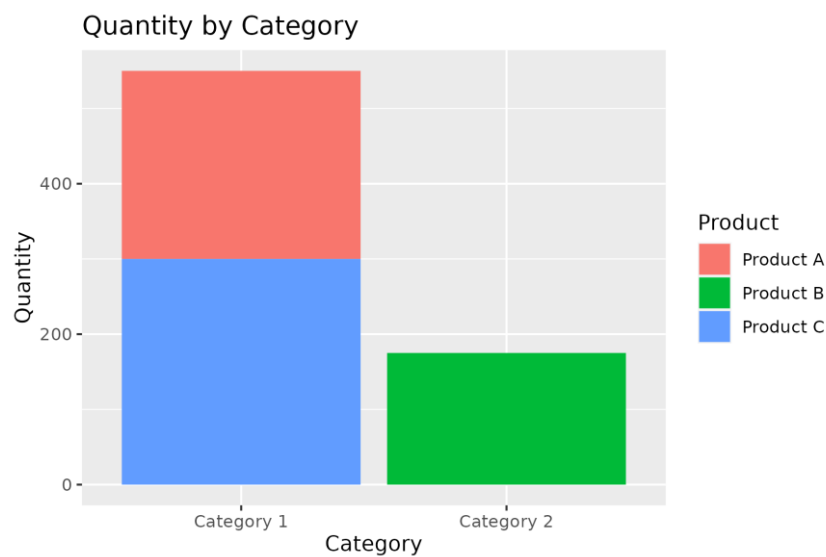
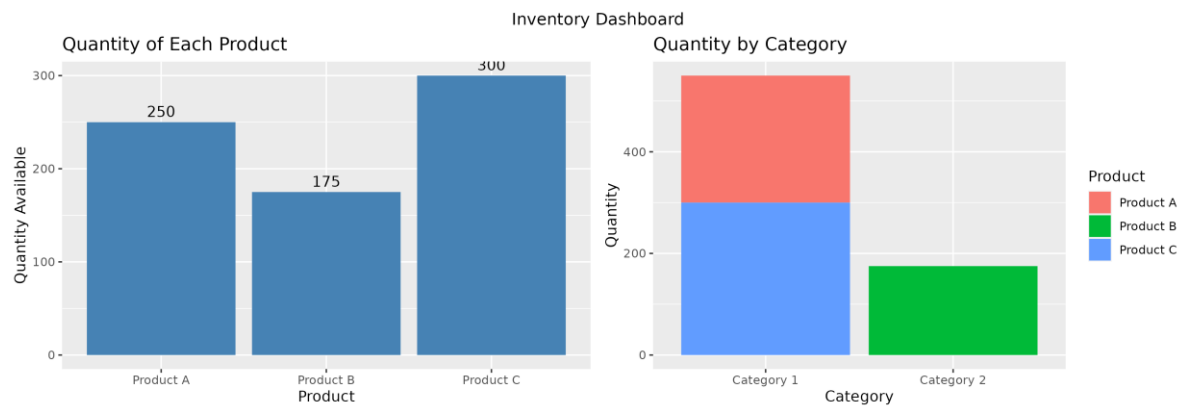


Figure 18.2: Q2 - Stacked bar chart by category (assumed)

Product Inventory Table

Product_ID	Product	Quantity	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Figure 18.3: Q3 - Inventory table



Product_ID	Product	Quantity	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Figure 18.4: Q4 - Dashboard

Result

- Product C has the most stock (300) and Product B the least (175).
- Total stock is 725 units.
- Categories are assumed, so the stacked chart is for demonstration.

Set 19: Survey Responses Analysis

Data Visualization using R

Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

8. Open RStudio and load the packages `ggplot2` and `gridExtra` (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
9. Create the survey data as a data frame.
10. Question 1: draw the grouped bar chart of Question 1 answers with `geom_bar(position = "dodge")`.
11. Question 2: draw the stacked bar chart of all three questions with `geom_bar()`.
12. Question 3: make the table with `tableGrob()`.
13. Question 4: join the two charts and the table into a dashboard with `arrangeGrob()`.
14. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
survey <- data.frame(Survey_ID = 1:3, Question1 = c("A","B","C"),
                     Question2 = c("B","A","A"), Question3 = c("C","D","B"))
print(survey)
long <- stack(survey[, 2:4]); names(long) <- c("Answer", "Question")
print(table(long$Question, long$Answer))

# 1. Grouped bar chart - Question 1
p1 <- ggplot(survey, aes(Question1, fill = factor(Survey_ID))) +
  geom_bar(position = "dodge") +
  labs(title = "Answers of Question 1", x = "Answer", y = "Count", fill = "Survey ID")

# 2. Stacked bar chart - all questions
p2 <- ggplot(long, aes(Question, fill = Answer)) + geom_bar() +
  labs(title = "Responses of All Questions", x = "Question", y = "Count")

# 3. Table
tb <- tableGrob(survey, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Survey Dashboard")

ggsave("set19_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set19_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set19_q3.png", arrangeGrob(tb, top = "Survey Response Table"), width = 5, height = 2, bg = "white")
ggsave("set19_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
Survey_ID Question1 Question2 Question3
1      1      A      B      C
2      2      B      A      D
3      3      C      A      B

  A B C D
Question1 1 1 1 0
Question2 2 1 0 0
```

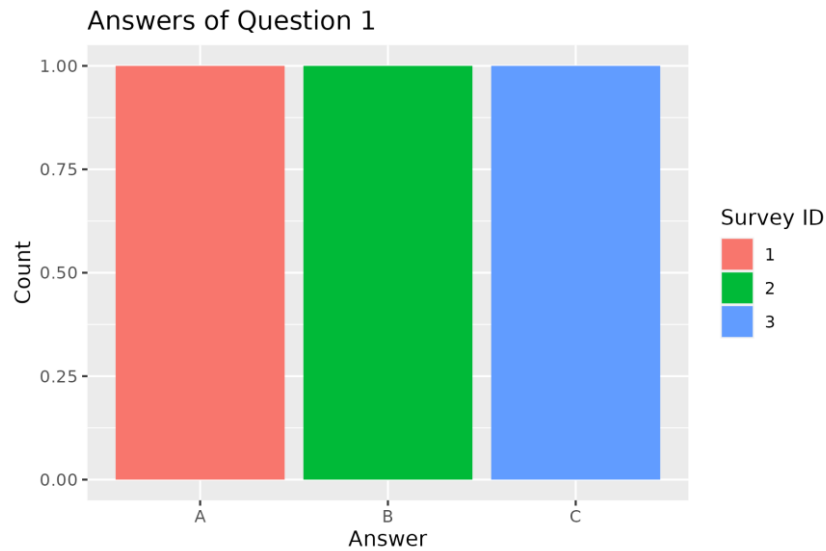


Figure 19.1: Q1 - Grouped bar chart (Question 1)

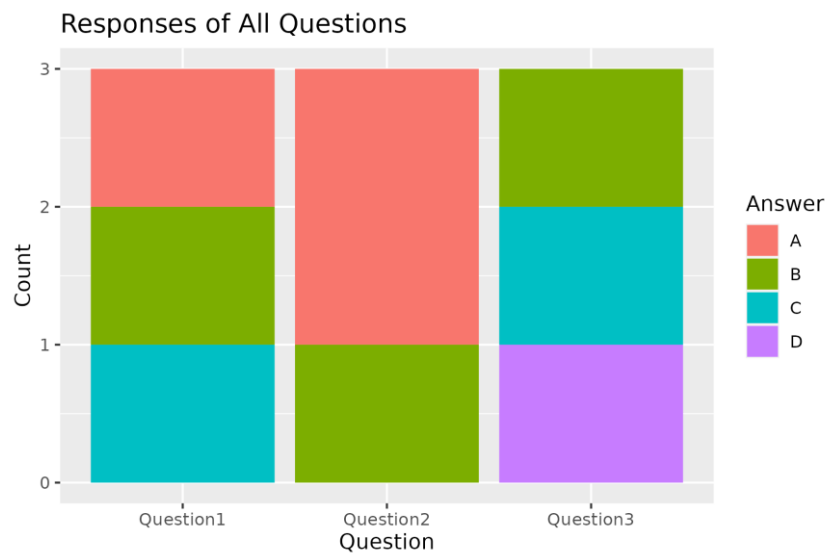
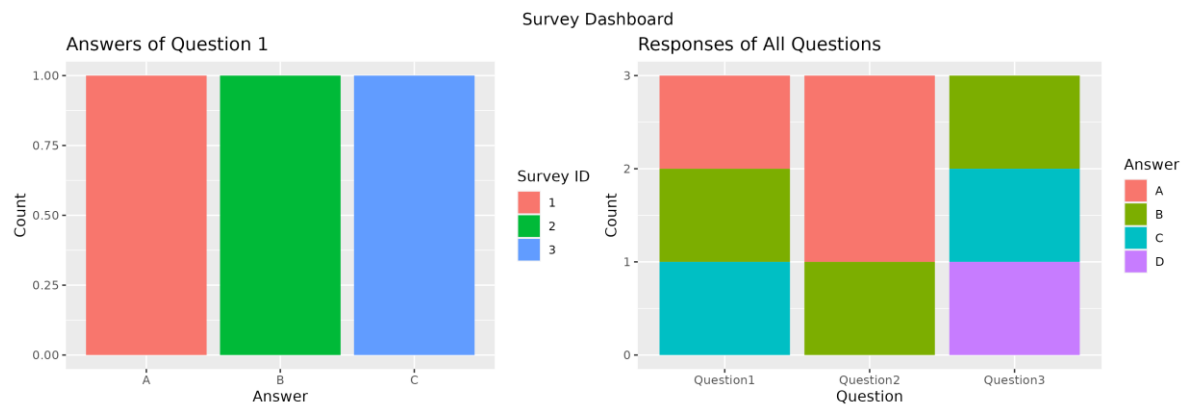


Figure 19.2: Q2 - Stacked bar chart (all questions)

Survey Response Table

Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 19.3: Q3 - Survey response table



Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 19.4: Q4 - Dashboard

Result

- In Question 1, answers A, B and C were chosen once each.
- Question 2 has A twice and B once; Question 3 has B, C and D once each.
- All four parts were completed successfully in R.

Set 20: Stock Analysis

Data Visualization using R

Aim

To analyse stock prices using a line chart, a bar chart, a table and a dashboard in R.

Procedure

15. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
16. Create the stock data as a data frame.
17. Question 1: draw the line chart of the three stocks with `geom_line()`.
18. Question 2: calculate the daily % change of Stock A with `diff()` and draw the bar chart.
19. Question 3: make the table with `tableGrob()`.
20. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
21. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
stocks <- data.frame(Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
                     Stock_A = c(100, 105, 110), Stock_B = c(150, 152, 148),
                     Stock_C = c(120, 118, 122))
print(stocks)
change <- data.frame(Date = stocks$Date[-1],
                     Change = diff(stocks$Stock_A) / head(stocks$Stock_A, -1) * 100)
print(change)

# 1. Line chart - three stocks
p1 <- ggplot(stocks, aes(Date)) +
  geom_line(aes(y = Stock_A, colour = "Stock A")) +
  geom_line(aes(y = Stock_B, colour = "Stock B")) +
  geom_line(aes(y = Stock_C, colour = "Stock C")) +
  labs(title = "Stock Prices", x = "Date", y = "Price", colour = "Stock")

# 2. Bar chart - daily % change of Stock A
p2 <- ggplot(change, aes(factor(Date), Change)) + geom_col(fill = "seagreen") +
  geom_text(aes(label = paste0(round(Change, 2), "%")), vjust = -0.5) +
  labs(title = "Daily % Change of Stock A", x = "Date", y = "Change (%)")

# 3. Table
tb <- tableGrob(stocks, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Stock Dashboard")

ggsave("set20_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set20_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set20_q3.png", arrangeGrob(tb, top = "Stock Price Table"), width = 5, height = 2, bg = "white")
ggsave("set20_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Date	Stock_A	Stock_B	Stock_C
1	2023-01-01	100	150	120
2	2023-01-02	105	152	118
3	2023-01-03	110	148	122

Date	Change
2023-01-02	5%
2023-01-03	10%

1 2023-01-02 5.000000
2 2023-01-03 4.761905

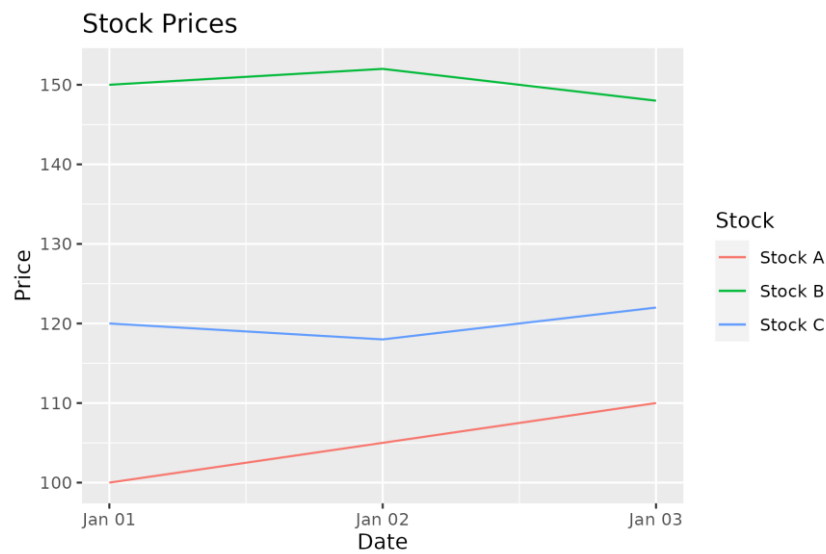


Figure 20.1: Q1 - Stock price line chart

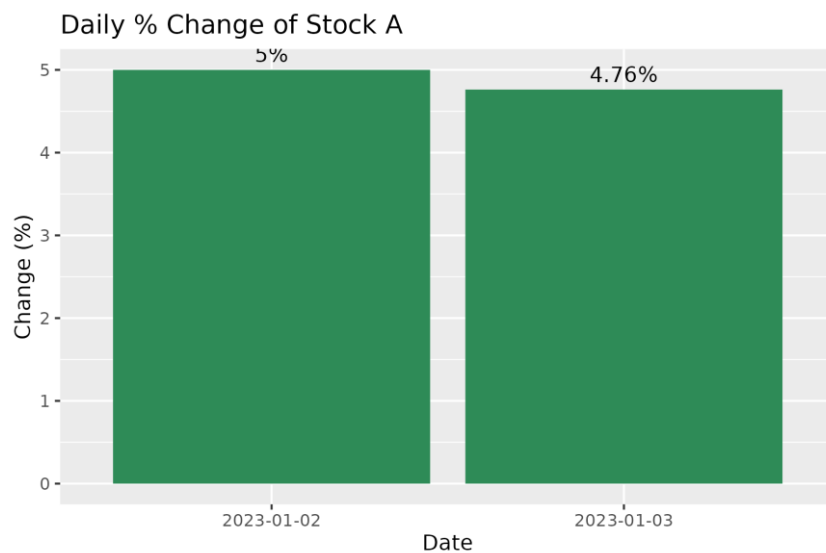


Figure 20.2: Q2 - Daily % change of Stock A

Stock Price Table

Date	Stock_A	Stock_B	Stock_C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Figure 20.3: Q3 - Stock price table

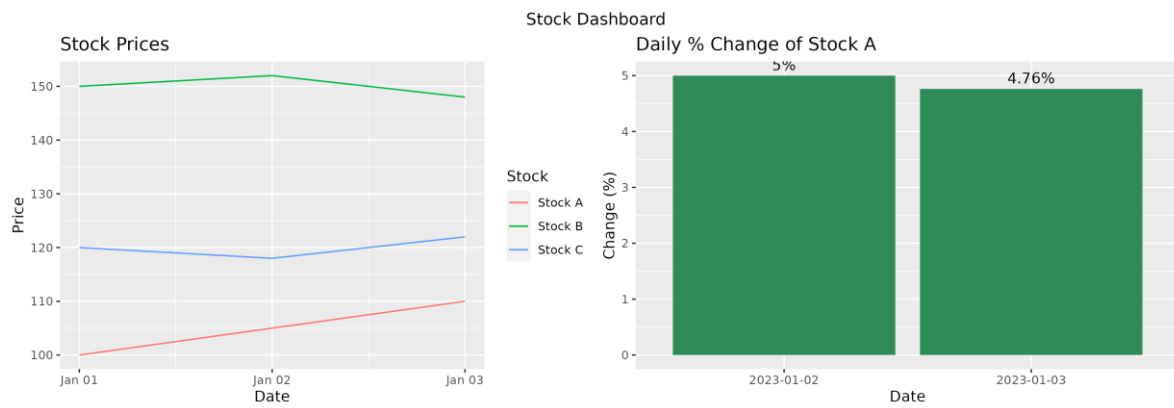


Figure 20.4: Q4 - Dashboard

Result

- Stock A rose from 100 to 110, with daily changes of +5.00% and +4.76%.
- Stock B and Stock C moved up and down.
- All four parts were completed successfully in R.

Set 21: Energy Consumption Analysis

Data Visualization using R

Aim

To analyse energy consumption data in R using a histogram, a density plot, a bubble scatter plot, a bar chart and a dashboard with filters.

Procedure

22. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
23. Save the data as `Energy_Consumption_Data.csv` and import it with `read.csv()`.
24. Question 1: plot the histogram with `geom_histogram()` and the density curve with `geom_density()`.
25. Question 2: draw the scatter plot of temperature against units with bubble size = `Peak_Hours` and alpha for transparency.
26. Question 3: find the average `Renewable_Usage` of each sector with `aggregate()` and draw the bar chart.
27. Question 4: draw the monthly line chart and the Sector vs Region heatmap, join them into a dashboard, and apply the Month and Sector filters with `subset()`.
28. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)
options(width = 95)

# Create the CSV file and import it
data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
) -> d
write.csv(d, "Energy_Consumption_Data.csv", row.names = FALSE)
energy <- read.csv("Energy_Consumption_Data.csv")
energy$Month <- factor(energy$Month, levels = c("Jan", "Feb", "Mar"))
print(energy)

# 1. Histogram and density plot
print(aggregate(Units_Consumed ~ Sector, energy, mean))
p1 <- ggplot(energy, aes(Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 150, fill = "skyblue", colour = "white") +
  geom_density(colour = "red") +
  labs(title = "Histogram and Density of Units Consumed", x = "Units Consumed (kWh)", y = "Density")

# 2. Scatter plot with bubble size and transparency
cat("Correlation:", cor(energy$Temperature, energy$Units_Consumed), "\n")
p2 <- ggplot(energy, aes(Temperature, Units_Consumed, size = Peak_Hours)) +
  geom_point(alpha = 0.5, colour = "blue") +
  labs(title = "Temperature vs Units Consumed", x = "Temperature (C)", y = "Units Consumed (kWh)")

# 3. Average renewable usage by sector
avg <- aggregate(Renewable_Usage ~ Sector, energy, mean)
print(avg)
```

```

p3 <- ggplot(avg, aes(Sector, Renewable_Usage, fill = Sector)) + geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0(Renewable_Usage, "%")), vjust = -0.5) +
  labs(title = "Average Renewable Usage by Sector", y = "Renewable Usage (%)")

# 4. Dashboard (line chart + heatmap); filters are applied with subset()
dash <- function(d, title) {
  line <- ggplot(aggregate(Units_Consumed ~ Month, d, sum), aes(Month, Units_Consumed, group = 1)) +
    geom_line() + geom_point(size = 3) + labs(title = "Monthly Energy Usage", y = "Units (kWh)")
  heat <- ggplot(d, aes(Region, Sector, fill = Units_Consumed)) + geom_tile() +
    geom_text(aes(label = Units_Consumed)) + scale_fill_gradient(low = "yellow", high = "red") +
    labs(title = "Heatmap: Sector vs Region", fill = "Units")
  arrangeGrob(line, heat, ncol = 2, top = title)
}
d1 <- dash(energy, "Energy Dashboard (No Filter)")
d2 <- dash(subset(energy, Month %in% c("Jan", "Feb") & Sector != "Industrial"),
  "Energy Dashboard (Filter: Jan-Feb, Residential and Commercial)")

ggsave("set21_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set21_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set21_q3.png", p3, width = 6, height = 4, bg = "white")
ggsave("set21_q4a.png", d1, width = 12, height = 5, bg = "white")
ggsave("set21_q4b.png", d2, width = 12, height = 5, bg = "white")

```

Output

	Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage	Peak	Hours
1	Residential	North	Jan	15	320	2100	22	4	
2	Commercial	South	Jan	24	540	3600	18	6	
3	Industrial	West	Feb	20	880	5900	12	8	
4	Residential	East	Feb	18	350	2300	25	5	
5	Commercial	North	Mar	28	610	4100	20	7	
6	Industrial	South	Mar	30	920	6200	15	9	

	Sector	Units_Consumed
1	Commercial	575
2	Industrial	900
3	Residential	335

Correlation: 0.654655

	Sector	Renewable_Usage
1	Commercial	19.0
2	Industrial	13.5
3	Residential	23.5

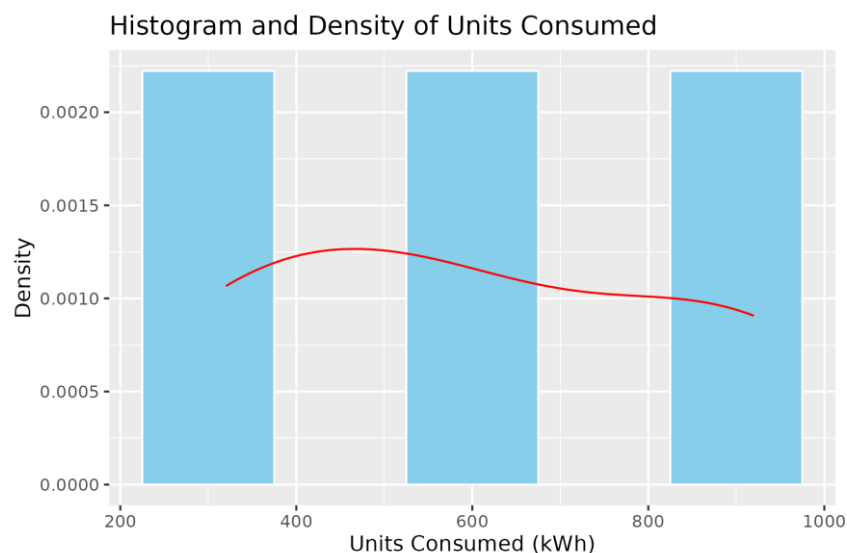


Figure 21.1: Q1 - Histogram and density plot

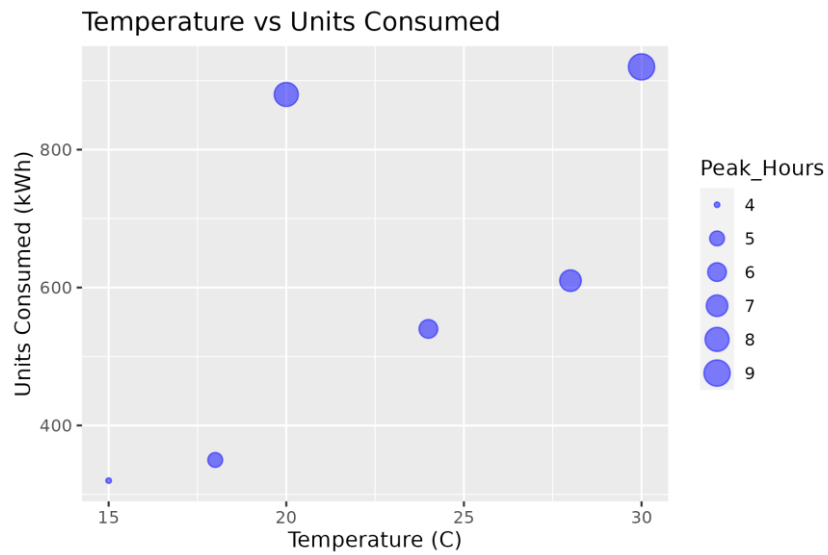


Figure 21.2: Q2 - Temperature vs units (bubble = peak hours)

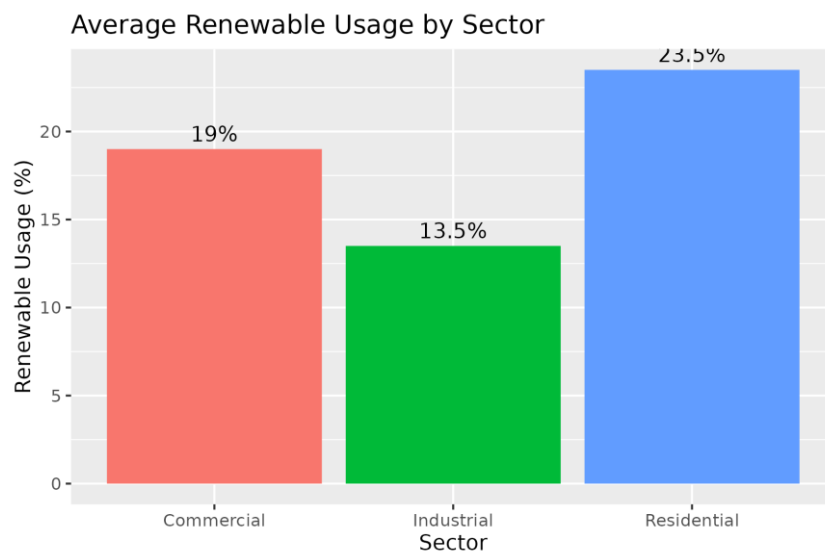


Figure 21.3: Q3 - Average renewable usage by sector

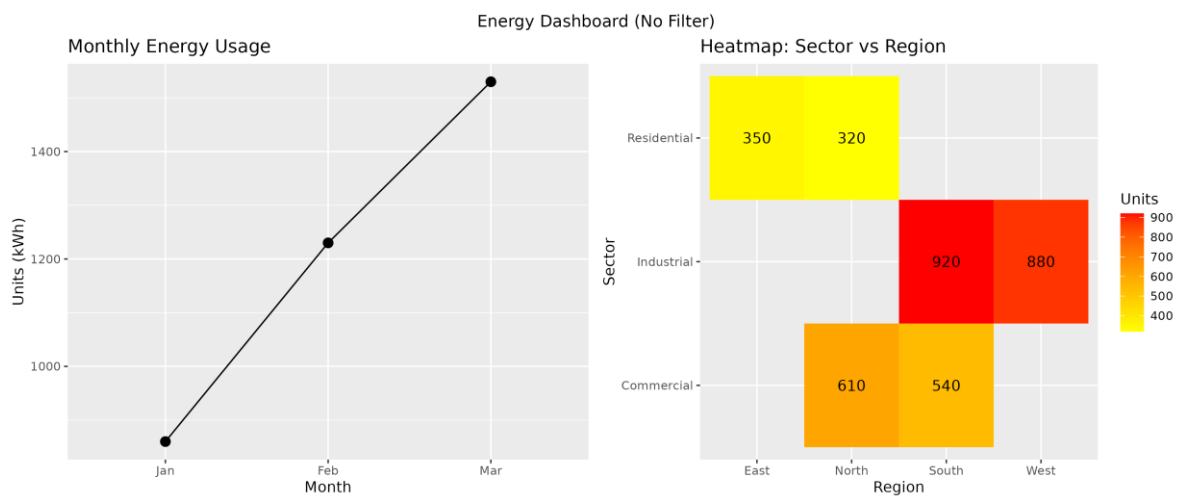


Figure 21.4: Q4 - Dashboard (no filter)

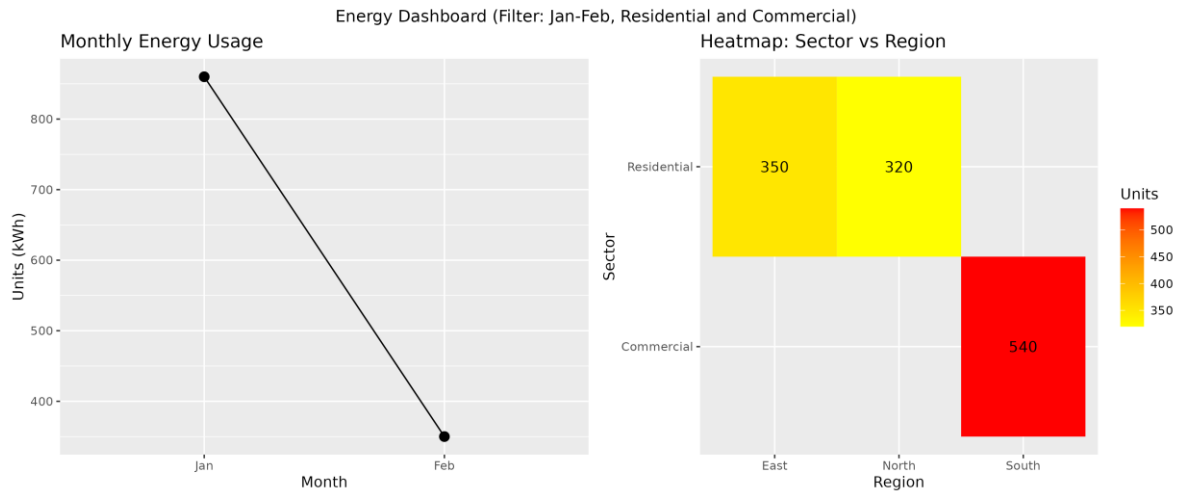


Figure 21.5: Q4 - Dashboard (filter: Jan-Feb, Residential and Commercial)

Result

- Industrial uses the most energy (average 900 kWh) and Residential the least (335 kWh).
- Temperature and units have a positive correlation of 0.65.
- Residential has the highest renewable usage (23.5%) and Industrial the lowest (13.5%), so Industrial has the most scope to improve.
- Energy use rises from January to March, and the heatmap shows the highest use in Industrial.